

Fabio A. CRUZ SANCHEZ

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CNU 60, 62

Fabrication Aditiva; Open Source; Reciclaje Distribuido; Sostenibilidad



Presentación

Profesor en la [École Nationale Supérieure en Génie des Systèmes et de l'Innovation –ENSGSI-](#), adscrita al laboratorio de investigación [Équipe de Recherche sur les Processus Innovatifs – ERPI-](#) de la Université de Lorraine en Francia. Ingeniero mecánico formado en la Universidad Nacional de Colombia, poseo un Máster II en Gestión de la Innovación y Diseño Industrial, así como un doctorado en Ingeniería de Sistemas Industriales por la Université de Lorraine. Mi área de investigación se centra en la fabricación aditiva de código abierto como palanca para el desarrollo industrial sostenible. Más concretamente, estoy estudiando nuevos canales de distribución cortos para la recuperación de plásticos reciclados mediante la fabricación aditiva. Adopto un enfoque multiescala para abordar las cuestiones tecnológicas, industriales y territoriales asociadas, al tiempo que favorezco la colaboración con diversas partes interesadas. Utilizo distintas metodologías para mejorar, probar y experimentar nuevos usos. Tengo una sólida experiencia en la dirección de proyectos de investigación (EU H2020, ANR) y de formación (ERASMUS+).

Formación

2013 – 2016

Ph.D., Université de Lorraine, Especialidad Ingeniería de Sistemas Industriales
 Título de Tesis: *Methodological proposition to evaluate polymer recycling in open-source additive manufacturing contexts*

Defendida públicamente el 9 de diciembre de 2016 en Nancy ante el jurado:

Rapporteurs:

- Prof. Nicolas PERRY – ENSAM, Bordeaux - France
- Dr. Salim BELOUETTAR – LIST, Esch-sur-Alzette - Luxembourg

Examineurs:

- Prof. Joshua M. PEARCE – MTU, Michigan - USA
- Prof. Nadia BAHLOULI – Université de Strasbourg, Strasbourg - France
- Prof. Mauricio CAMARGO (*Directeur*) – UL, ERPI, Nancy - France
- MdC. Hakim BOUDAUD (*co-directeur*) – UL ERPI, Nancy - France
- Dr. Sandrine HOPPE (*co-directeur*) – LRGP, Nancy - France

2012 – 2013

Master II. Management de l'Innovation et du Desing Industriel, Université de Lorraine, FR

Tesis: *Proposition d'un Protocole d'expérimentation standard pour la fabrication additive open source*

2004 – 2012

B.Sc. Ingénieur Mécanique, Universidad Nacional de Colombia, Bogotá, Colombia

Experiencia profesional

- 2024 – ... **Profesor Asistente** *Université de Lorraine*, Nancy – France
- 2022 – 2023 **Investigador** *Université de Lorraine*, Nancy – France
- 2021 – 2021 **Investigador** *Université de Technologie de Troyes*, Équipe InSyTe (Antiguamente CREIDD) Troyes – France
- 2017 – 2021 **Post-doctorante** *Laboratoire ERPI*, Université de Lorraine, Nancy – France
- 2010 – 2011 **International trainee** *Empresa MIP Engenharia S/A* Belo Horizonte, Brazil
Proyecto: Ayudar a crear un plan estratégico para el proyecto de internacionalización del PIM.
- 2008 – 2009 **Estudiante auxiliar de ingeniería** *Universidad Nacional de Colombia*. Bogotá - Colombia
Projet: Diseño y construcción de una máquina de fundición centrífuga para fabricar cilindros de fundición ASTM 40.

Competencias

-   Lengua materna  C2  C1  B2
-  CAO (Solid-Works, Onshape), Matlab, Data analysis / visualización, R, HTML, CSS
-  Investigación, Zotero, L^AT_EX and Rmarkdown/Quarto publishing.

Liste de la Production Scientifique

Articles à comité de lecture dans des revues internationales

1. Romani, A., Bogoya, A.B., Cruz Sanchez, F.A., Boudaoud, H., Camargo, M., Pearce, J.M., 2026. Open source technology readiness level assessment and application to the case study of distributed recycling for additive manufacturing. *Technological Forecasting and Social Change* 229, 124700. <https://doi.org/10.1016/j.techfore.2026.124700>
2. Suescun Gonzalez, C., Sandei, B., **Cruz Sanchez, F.A.**, Hoppe, S., Boudaoud, H., Pearce, J.M., Nouvel, C., 2026. Compatibilizer effectiveness for the reuse of mixed post-consumer solid waste plastic towards Distributed recycling additive manufacturing. *Materials Today Sustainability* 34, 101323. <https://doi.org/10.1016/j.mtsust.2026.101323>
3. Gonzalez, C.S., Basdeo, A., **Cruz Sanchez, F.A.**, Nouvel, C., Pearce, J.M., Boudaoud, H., 2025. Strategies for recycling multi-material polymer blends for additive manufacturing. *Sustainable Materials and Technologies* 44, e01430. <https://doi.org/10.1016/j.susmat.2025.e01430>
4. Liu, D., Basdeo, A., Gonzalez, C.S., Romani, A., Boudaoud, H., Nouvel, C., **Cruz Sanchez, F.A.**, Pearce, J.M., 2024. Low-Cost Open-Source Melt Flow Index System for Distributed Recycling and Additive Manufacturing. *Materials* 17, 5966. <https://doi.org/10.3390/ma17235966>
5. Suescun Gonzalez, C., **Cruz Sanchez, F.A.**, Boudaoud, H., Nouvel, C., Pearce, J.M., 2024. Multi-material distributed recycling via material extrusion: recycled high density polyethylene and poly (ethylene terephthalate) mixture. *Polymer Engineering & Science*. <https://doi.org/10.1002/pen.26643>
6. Osorio, F., **Cruz, F.**, Camargo, M., Dupont, L., Peña, J.I., 2024. Exploring team roles for social innovation labs: Toward a competence-based role self-assessment approach. *Journal of Engineering and Technology Management* 71, 101799. <https://doi.org/10.1016/j.jengtecman.2024.101799>
7. Caceres-Mendoza, C., Santander-Tapia, P., **Cruz Sanchez, F.A.**, Troussier, N., Camargo, M., Boudaoud, H., 2023. Life cycle assessment of filament production in distributed plastic recycling via additive

- manufacturing. *Cleaner Waste Systems* 5, 100100. <https://doi.org/10.1016/j.clwas.2023.100100>
8. Xue, F., Boudaoud, H., Robin, G., A. **Cruz Sanchez, F.**, Daya, E.M., 2023. Enhancing vibration damping properties of additively manufactured viscoelastic structures through process parameter optimization. *Mechanics of Advanced Materials and Structures* 0, 1–11. <https://doi.org/10.1080/15376494.2023.2245827>
 9. Gabriel, A., **Cruz, F.**, 2023. Open source IoT-based collection bin applied to local plastic recycling. *HardwareX* 13, e00389. <https://doi.org/10.1016/j.ohx.2022.e00389>
 10. López, V.M., Carou, D., **Cruz S, F.A.**, 2022. Feasibility study on the use of recycled materials for prototyping purposes: A comparative study based on the tensile strength. *Proceedings of the Institution of Mechanical Engineers, Part B: Journal of Engineering Manufacture* 09544054221113378. <https://doi.org/10.1177/09544054221113378>
 11. Xue, F., Robin, G., Boudaoud, H., **Cruz Sanchez, F.A.**, Daya, E.M., 2022. General Methodology to Investigate the Effect of Process Parameters on the Vibration Properties of Structures Produced by Additive Manufacturing Using Fused Filament Fabrication. *JOM* 74, 1166–1175. <https://doi.org/10.1007/s11837-021-05051-9>
 12. Santander, P., **Cruz Sanchez, F.A.**, Boudaoud, H., Camargo, M., 2022. Social, political, and technological dimensions of the sustainability evaluation of a recycling network. A literature review. *Clean. Eng. Technol.* <https://doi.org/10.1016/j.clet.2022.100397>
 13. Alexandre, A., **Cruz Sanchez, F.A.**, Boudaoud, H., Camargo, M., Pearce, J.M., 2020. Mechanical Properties of Direct Waste Printing of Polylactic Acid with Universal Pellets Extruder: Comparison to Fused Filament Fabrication on Open-Source Desktop Three-Dimensional Printers. *3D Print. Addit. Manuf.* 3dp.2019.0195. <https://doi.org/10.1089/3dp.2019.0195>
 14. Santander, P., **Cruz Sanchez, F.A.**, Boudaoud, H., Camargo, M., 2020. Closed loop supply chain network for local and distributed plastic recycling for 3D printing: a MILP-based optimization approach. *Resour. Conserv. Recycl.* 154, 104531. <https://doi.org/10.1016/j.resconrec.2019.104531>
 15. **Cruz Sanchez, F.A.**, Boudaoud, H., Camargo, M., Pearce, J.M., 2020. Plastic recycling in additive manufacturing: A systematic literature review and opportunities for the circular economy. *J. Clean. Prod.* 264, 121602. <https://doi.org/10.1016/j.jclepro.2020.121602>
 16. **Cruz Sanchez, F.A.**, Boudaoud, H., Hoppe, S., Camargo, M., 2017. Polymer recycling in an open-source additive manufacturing context: Mechanical issues. *Addit. Manuf.* 17, 87–105. <https://doi.org/10.1016/j.addma.2017.05.013>
 17. **Cruz Sanchez, F.A.**, Boudaoud, H., Muller, L., Camargo, M., 2014. Towards a standard experimental protocol for open source additive manufacturing. *Virtual Phys. Prototyp.* 9, 151–167. <https://doi.org/10.1080/17452759.2014.919553>

Communications dans des congrès internationaux à comité de lecture

1. Xue, F., Boudaoud, H., Robin, G., **Cruz Sanchez, F. A.**, & Daya, E. M. (2022). Influence of layer thickness and nozzle temperature on the interlocking adhesion strength of additive manufactured multi-material interface. In *2022 International Solid Freeform Fabrication Symposium*.
2. Xue, F., Robin, G., Boudaoud, H., **Cruz Sanchez, F. A.**, & Daya, E. M. (2021). Effect of process parameters on the vibration properties of PLA structure fabricated by additive manufacturing. In *2021 International Solid Freeform Fabrication Symposium*. University of Texas at Austin.
3. Roux-Marchand, T., **Cruz, F.**, Dupont, L., Camargo, M., Osorio, F., 2020. Connecting the strategic intent of innovation labs and projects: the case of the Green Fablab, in: *2020 IEEE*

International Conference on Engineering, Technology and Innovation (ICE/ITMC). IEEE, pp. 1–10. <https://doi.org/10.1109/ICE/ITMC49519.2020.9198320>

4. Pavlo, S., **Fabio, C.**, Hakim, B., Mauricio, C., 2018. 3D-Printing Based Distributed Plastic Recycling: A Conceptual Model for Closed-Loop Supply Chain Design, in: 2018 IEEE International Conference on Engineering, Technology and Innovation (ICE/ITMC). IEEE, pp. 1–8. <https://doi.org/10.1109/ICE.2018.8436296>
5. Albuquerque, R., Arbelaez, G., **Cruz, F.**, Camargo, M., Joseph, D., Tran, N., 2018. Modelling, Printing and Validation of Dental Dry Models for Implantology Skills Training, in: 2018 IEEE International Conference on Engineering, Technology and Innovation (ICE/ITMC). IEEE, pp. 1–8. <https://doi.org/10.1109/ICE.2018.8436302>
6. **Cruz, F.**, Lanza, S., Boudaoud, H., Hoppe, S., Camargo, M., 2015. Polymer Recycling and Additive Manufacturing in an Open Source context : Optimization of processes and methods, in: Solid Freeform Fabrication. Austin, Texas, pp. 1591–1600.

Chapitre d'Ouvrages scientifiques

1. Mendoza, C.C., Tapia, P.S., **Cruz, F.**, Troussier, N., Camargo, M., Boudaoud, H., 2024. *Chapitre 5 : Fabrication additive : le recyclage local de plastique*, in: Territoires en transition et innovation: Enjeux et outils d'accompagnement. p. 137.
2. **Cruz, F.**, Lanza, S., Boudaoud, H., Hoppe, S., Camargo, M., “*Approche méthodologique pour évaluer la faisabilité du recyclage des polymères dans un contexte de fabrication additive open-source*” (2023). Le recyclage, enjeu pour l'économie circulaire. ISTE Group
3. Pavlo, S., **Fabio, C.**, Hakim, B., Mauricio, C., “*Recyclage distribué en circuit court de plastiques revalorisés par l'impression 3D « Open Source » : Un cadre pour l'analyse et l'évaluation de sa durabilité*” (2023). Le recyclage, enjeu pour l'économie circulaire. ISTE Group

Colloques et congrès nationales

1. Marche, B., Kasmi, F., **Sanchez, F.**, Mayer, F., & Dupont, L. (2020, March). From technological demonstrator to sustainable supply chain demonstrator: a conceptual framework. In Spring of Innovation.
2. **Fabio Cruz**, Hakim Boudaoud, Sandrine Hoppe, Mauricio Camargo. Le concept du Green Fablab: une approche distribuée pour le recyclage de polymères en circuit-court dans le cadre de la fabrication additive open-source. 15ème Congrès de la Société Française de Génie des Procédés (SFGP), Jul 2017, Nancy, France. hal-02518997

Encadrement Doctoral et Scientifique

Au niveau Master

- (2025) Nicolas Carcamo: *Assessing the Sustainability of the Plastics Industry: A System Dynamics and Optimization Approach*. 100%
- (2026) Rodrigo Contrera Nuñez: *Techno-ecological synergies for a post-growth distributed recycling context: A complementary approach of Life cycle assessments and ecosystem services*. 50%
- (2027) Alfonso Enrique Suarez: *Proposal for a business model for the management of household plastic*. 50%
- (2028) Ignacio Castaneda: *Proposal of a local distributed recycling optimization model: application to plastics and metals obtained from e-waste*. 50%

- (2029) Cristian Caceres: *Recyclage distribué des déchets plastiques et rôle de la fabrication additive : proposition d'un modèle d'affaire économique et environnemental*. 50%
- (2030) Swapnil Balpande: *Quality in Fused Granular Deposition for distributed recycling*. 50%
- (2031) Anamaria Barrera Bogoya: *A smart grid of plastics – Technical conditions for the sorting and pre-step evaluation for a “factory of the future”*. 50%
- (2032) Arthur Alexandre: *Mechanical Properties of Direct Waste Printing of Polylactic Acid with Universal Pellets Extruder: Comparison to Fused Filament Fabrication on Open-Source Desktop Three-Dimensional Printers*. 50%
- (2033) Pavlo Santander: *3D-Printing based distributed plastic recycling: A conceptual model for closed-loop supply chain design* 50%
- (2034) Ramsés Albuquerque: *Modelling, printing and validation of dental dry models for implantology skills training* 50%
- (2035) Silvia Lanza: *Polymer Recycling and Additive Manufacturing in an Open Source context: Optimization of processes and methods* 50%